AI Generated Text and its Effect on Text Generation

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Abstract

Recursive training, an issue in training where language models are trained on text generated by a language model, is a critical risk in the field of NLP. Where most studies focus on improvement of Large Language Models (LLMs), the issue of model collapse is less focused. This project aims to study the impact of recursive training on n-gram language models and Recurrent Neural Networks. The results of this study seek to provide a quantifiable metric of the erosion of these language models during recursive training.

1 Motivation

The rise of AI generated text in online forums and social media has led to an unprecedented rise in artificial text in many data repositories. This rise presents a new risk known as "model collapse," which occurs when a model is trained on its own output and the model's output degrades. Model Collapse is a very prevalent issue for models trained on freshly scraped web data.

A substantial portion of current research focuses on the power and reliability of Transformer-based architecture, the mechanics of this degradation remain hard to find. I propose an investigation into n-gram models and Recurrent Neural Networks (RNNs) because their structure proves very vulnerable to model collapse. Unlike LLMs, which hide these issues by the large number of parameters, n-grams and RNNs provide a clear window through which to observe the degradation of language models on a small scale.

2 Problem Statement

2.1 Formal Definition of Data Mixing

Let D_H represent a corpus of human writing Amazon reviews and D_A represent a corpus of reviews generated by a Language Model. We define the training set D_{train} as a mixture of:

$$D_{train} = (1 - \alpha)D_H + \alpha D_A$$

where $\alpha \in [0,1]$ represents the artificial contamination ratio. The objective is to evaluate how the model M deviates from the human baseline as α increases.

2.2 Architectural Comparison

The study evaluates two distinct language model architectures to compare their resilience to artificial noise:

- n-gram Models: A statistical approach where $P(w_i)$ depends on the history of n – 1 tokens. This study will observe the diversity of vocabulary and collapse of less common text sequences.
- Recurrent Neural Networks: A neural approach which maintains a hidden state h_i to understand text context. This study will observe if this approach is less prone to model collapse compared to the statistical models.

2.3 Research Objective

The core aim of this study is to identify the threshold of α at which:

- Vocabulary Complexity Collapse: Defined by a significant reduction in the unique tokens and Shannon Entropy of generated examples

- Human Detection Failure: Defined by a point where human evaluators no longer feel confident in distinguishing between generated text and human text.

3 Proposed Model/Algorithm

3.1 n-gram Setup

This study will use a 5-gram language model to serve as the statistical model baseline. Kneser-Ney smoothing will also be applied to manage the sparsity of the 5-word context window present.

Text will be generated using Top-k sampling to prevent the model from generating in textual loops.

3.2 RNN Setup

This study will use a Long Short-Tem Memory (LSTM) network. The usage of a LSTM will soothe the issue that vanilla RNNs have when processing long responses.

4 Methodology

4.1 Data Mixing and Contamination

The core experiment will be conducted across five distinct phases, where the "artificial contamination ratio" α increases by 25% in each step.

Phase	D_H	D_A	α
0 (Base)	80,000	0	0%
1	60,000	20,000	25%
2	40,000	40,000	50%
3	20,000	60,000	75%
4	0	80,000	100%

4.2 Recursive Generation Loop

To study the recursive effect, the artificial data (D_A) used in each phase will be recursive

- Phase 1: Uses 20k reviews generated by the Base model.
- Phase 2: Uses 40k reviews generated by the Phase 1 model.
- Phase 3: Uses 60k reviews generated by the Phase 2 model.
- Phase 4: Uses 80k reviews generated by the Phase 3 model.

This method ensures that errors and biases from previous generations are compounded, mimicking the real-world scenario of AI models learning from contaminated web data.

5 Datasets

5.1 Human Baseline

This study will use the Amazon Fine Food Reviews dataset (J. McAuley & J. Leskovec, 2013). This corpus contains 568,454 reviews.

5.2 Artificial Contamination

This study will create its own artificial reviews during each phase of model training and testing.

6 Measurement of Success

6.1 Intrinsic Metrics

This study will measure the unique words, the frequency of words, and the Shannon Entropy of a large amount of generated text at each of the 20k increments. We expect a noticeable decline in entropy as models begin to favor generic phrases over specific semantics.

6.2 Extrinsic Metrics

We will recruit N = 10 participants for each phase for a "blind" detection task.

Each participant will be shown 20 randomly selected reviews, 10 from the dataset and 10 artificially generated phrases.

Each participant will be asked to classify the review as either "Human" or "Artificial" and rate the confidence of their answer on a scale from 1 (Guessed) to 5 (Certain).

7 Limitations

7.1 Architectural Simplicity

This research focuses on n-gram models and LSTMs to ensure timely results and technical simplicity. However, these models lack the large parameter scales and attention mechanisms that modern Transformer-based LLMs used. Consequently, the specific decline identified in this

study may manifest at different points in LLMs than these models.

7.2 Domain Specificity

The use of a dataset of specifically food reviews may limit the variety of text which may not generalize to other domains such as creative writing or technical documentation.

8 Ethics Statement

8.1 Responsible Data Use

This study involves generation of synthetic amazon reviews. These reviews are strictly for research purposes. To prevent the spread of misinformation generated reviews will not be uploaded to any live consumer platforms and the synthetic corpus will be clearly labeled in any publicly shared repositories.

8.2 Human Participants

The human detection survey will be conducted under the following guidelines:

- Participants will be informed beforehand that they are participating in a study involving AI-generated text.
- No personally identifiable information (PII) will be collected during the survey process.
- Participants may exit the survey at any time without penalty, and their responses will be discarded.
- All generated text used in surveys will be scanned for offensive language.

References

Kneser R, Ney H. Improved backing-off for m-gram language modeling. IEEE International Conference on Acoustics, Speech, and Signal Processing; 1995.

J. McAuley and J. Leskovec. From amateurs to connoisseurs: modeling the evolution of user expertise through online reviews. WWW, 2013.